

# CAPSTONE – II

## PROGRESS REPORT

**Project Name:** Sentiment Analysis using Hierarchical Reasoning Models and Mixture-of-Experts

Architecture: Experimental Results and Analysis

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### Tasks Completed:

#### 1. Analysis of Model Failures & Performance Evaluation:

- Verified Traditional ML superiority over basic DL/Transformers in current settings
- Identified root causes for Transformer (frozen weights) and HRM underperformance.

#### 2. Domain-Specific Dataset Acquisition (4 Domains)

- Financial PhraseBank (Finance)
- HRAST (Hospitality)
- Medical Sentiment Analysis (Healthcare)
- Yelp Open Dataset (Local Business)

#### 3. Data Pipeline Implementation (New)

- Successfully implemented a robust preprocessing pipeline (dummy/preprocess.py) for all 9 acquired datasets.
- Standardized text input: Converted diverse formats (CSV, JSON, Latin-1) to UTF-8 and performed Regex cleaning for URLs, handles, and special characters.
- Normalized Labels: Created both Binary (Positive/Negative) and Ternary (Positive/Neutral/Negative) versions to address class imbalances and neutral sentiment handling.
- Validated dataset integrity across domains (Finance, Hospitality, Healthcare, Social Media).

#### 4. Model Updation

- Modified model code to handle multi-label data.
- Initialized training configurations for individual domains.

**Total Hours:** 52 hours (4-weeks)

### Plan for Next Week:

#### 1. Machine Learning Analysis Execution

- Execute ML baselines (Logistic Regression, Linear SVC, Naive Bayes, Random Forest).
- Evaluate performance metrics (Accuracy, F1, Precision, Recall) to establish benchmarks for domain-specific tasks.

#### 2. Deep Learning Models:

- Implement and fine-tune Transformers/HRM on the preprocessed data. Compare DL performance against the established ML baselines.